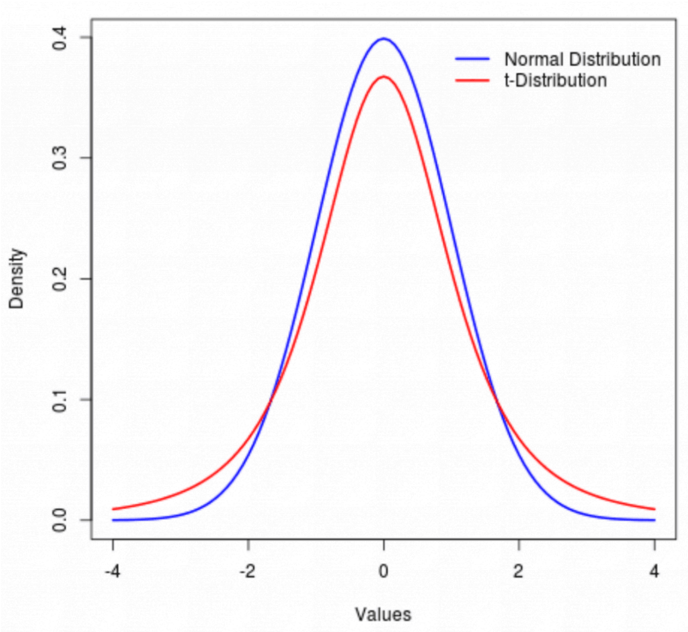
- Student's t-distribution, t-test, pairwise t-test.
- Hypothesis test, p-value, α, "confidence", type-I/II errors.
- F-statistic, the expectations in the book and its p-value
- AIC, BIC, Adjusted R2, Mallow's Cp - the tradeoffs between "model explanation and variable addition" that terms introduce in each of these statistics' formulae.
- synergy effect



$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

$$Z \sim \mathcal{N}(0, 1)$$

In reality:

- σ is unknown
- We estimate it using the **sample standard deviation**

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

Now substitute s instead of σ:

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

This quantity **does not follow a normal distribution.**

We **estimate noise variance**, and that propagates into the SE of β^

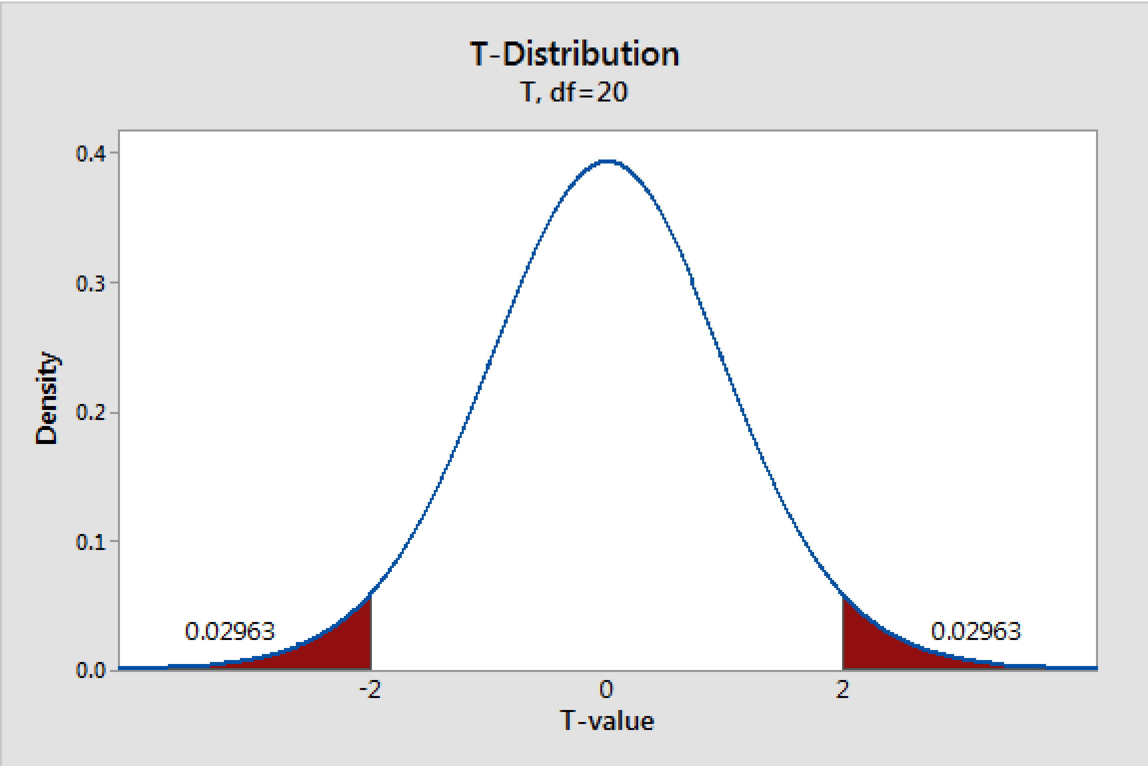
T Distribution Formula

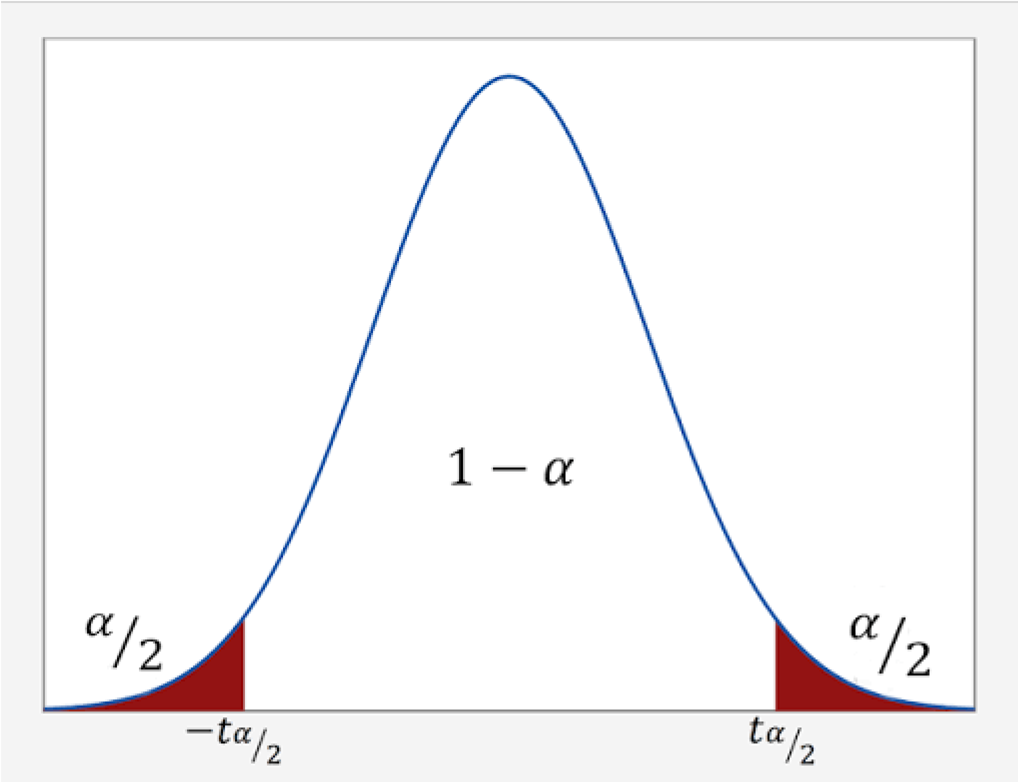
$$t = \frac{(x - \mu)}{\frac{s}{\sqrt{n}}}$$

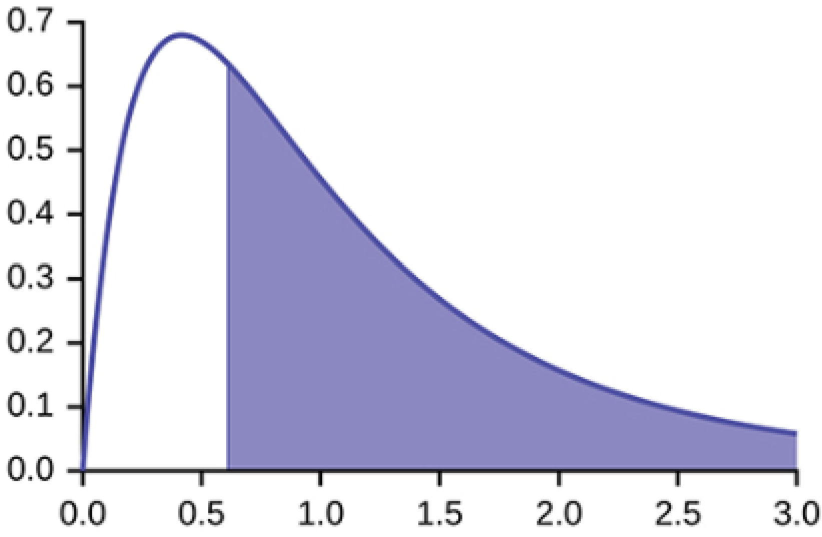
Paired Samples t-Test

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)},$$







$$C_p = \frac{1}{n}\left(\mathrm{RSS} + 2d\hat{\sigma}^2\right),$$

$$\text{AIC} = \frac{1}{n\hat{\sigma}^2}\left(\text{RSS} + 2d\hat{\sigma}^2\right),$$

$$\text{BIC} = \frac{1}{n\hat{\sigma}^2}\left(\text{RSS} + \log(n)d\hat{\sigma}^2\right).$$

$$\text{Adjusted } R^2 = 1 - \frac{\text{RSS}/(n-d-1)}{\text{TSS}/(n-1)}.$$

$$sales = \beta_0 + \beta_1 \cdot TV + \beta_2 \cdot radio + \beta_3 \cdot (TV \times radio) + \varepsilon$$

That new term, **β_3(TV × radio)**, is called the **interaction term**, and β_3 is the **interaction coefficient**.